IX.E. Cosmological Minisuperspace Extension: Scalar Field as a Quantum Clock for the Cosmic Scale Factor

Applying the **Hilbert-Space Quadrature Conjecture** to quantum cosmology resolves the historical "problem of time" in Wheeler--DeWitt minisuperspace models~\cite{DeWitt1967,Kuchar1992}. In a spatially homogeneous, isotropic flat Friedmann--Robertson--Walker (FRW) universe sourced by a homogeneous scalar field \varphi with potential V(\varphi), the universal state \vert{}\Psi\rangle depends on the scale factor a and the scalar field \varphi.
Whereas standard canonical quantum cosmology treats either a or \varphi as an ad-hoc external time coordinate, our framework establishes \varphi as an internal quantum clock sector (\mathcal{H}_C) whose circular phase amplitudes and quadrature distributions generate the emergent cosmological evolution of the scale factor (a \in \mathcal{H}_S).

1. The Minisuperspace Wheeler--DeWitt Constraint

The Arnowitt--Deser--Misner (ADM) canonical Hamiltonian constraint for a flat FRW universe (K=0) with scale factor a > 0 and scalar field \varphi \in (-\infty, \infty) takes the minisuperspace operator form (adopting Planck units \hbar = c = 1, with factor-ordering parameter p): \begin{equation} \widehat{H}_{\mathrm{WDW}} |\Psi\rangle = \left[\frac{1}{2m_0} a^{-p} \frac{\partial}{\partial a} \left(a^p \frac{\partial}{\partial a} \right) - \frac{1}{2a^3} \frac{\partial^2}{\partial \varphi^2} + a^3 V(\varphi) \right] |\Psi(a, \varphi)\rangle = 0, \label{eq:wdw_frw} \end{equation} where m_0 = \frac{3\pi}{2G} is the effective minisuperspace ``mass'' for the scale factor (proportional to the inverse Newton constant).
Notice the signature structure of the DeWitt metric on minisuperspace: the scale factor kinetic term has a negative sign (representing gravitational instability/hyperbolic expansion), while the scalar field kinetic term has a positive sign. Thus, \varphi acts naturally as a hyperbolic clock variable whose momentum drives the relational evolution of a.
To map this onto our circular phase framework, we introduce a canonical phase transformation on the scalar field clock sector, decomposing \varphi into an action-angle or polar clock representation (\rho, \theta_\varphi) where \theta_\varphi \in S^1 parameterizes cyclic scalar field phase oscillations, or equivalently treating the scalar field momentum \pi_\varphi = -i\frac{\partial}{\partial \varphi} as the clock generator \widehat{N}_\varphi.

2. Emergent Scale-Factor Schrödinger Equation and Conditional Friedmann Dynamics

Theorem 6 (Scalar-Field Clock Quadrature and Emergent Scale-Factor Evolution)

Let \vert{}\Psi(a, \varphi)\rangle be a physical minisuperspace state satisfying the Wheeler--DeWitt constraint Eq.~\eqref{eq:wdw_frw}. Define the scalar-field clock Hilbert space as \mathcal{H}_C = L^2(\mathbb{R}, d\varphi) (or its compactified periodic covering H^2(S^1)) with covariant phase/field POVM E(d\varphi) = \vert{}\varphi\rangle\langle\varphi\vert{}\,d\varphi, and the scale-factor system Hilbert space as \mathcal{H}_S = L^2(\mathbb{R}_+, a^p da). Then:

1. **Conditional Scale-Factor State:** *Conditioning the universal state on a scalar field clock reading \varphi yields the unnormalized conditional scale-factor wave function:* \begin{equation} |\psi(a | \varphi)\rangle := \langle \varphi | \Psi \rangle \in \mathcal{H}_S. \label{eq:cond_scale_factor} \end{equation}
2. **Emergent Cosmological Schr"odinger Equation:** *In regions where the scalar field potential V(\varphi) and minisuperspace kinetic terms vary adiabatically such that the effective scale-factor Hamiltonian \widehat{H}_{\mathrm{univ}}(a, \varphi) = \sqrt{2a^3 [a^3 V(\varphi) - \mathcal{E}]} is well-defined, the normalized conditional scale-factor state \rho_S(\varphi) = \vert{}\psi(a\vert{}\varphi)\rangle\langle\psi(a\vert{}\varphi)\vert{} / \Vert{}\psi(a\vert{}\varphi)\Vert{}^2 satisfies the emergent Tomonaga--Schwinger / Schr"odinger equation with respect to clock time \varphi:* \begin{equation} i \frac{\partial}{\partial \varphi} |\psi(a | \varphi)\rangle = \widehat{H}*{\mathrm{cosmo}}(a, \varphi) |\psi(a | \varphi)\rangle, \qquad \widehat{H}*{\mathrm{cosmo}}(a, \varphi) := \sqrt{\frac{2m_0}{a^3}} \left(a^3 V(\varphi) - \frac{1}{2m_0} a^{-p}\partial_a(a^p \partial_a)\right). \label{eq:cosmo_schrodinger} \end{equation}

Proof

Rewrite the Wheeler--DeWitt constraint Eq.~~\eqref{eq:wdw_frw} by isolating the scalar field momentum derivative: \begin{equation} -\frac{\partial^2}{\partial \varphi^2} |\Psi(a, \varphi)\rangle = 2a^3 \left(-\frac{1}{2m_0} a^{-p} \frac{\partial}{\partial a} \left(a^p \frac{\partial}{\partial a} \right) + a^3 V(\varphi) \right) |\Psi(a, \varphi)\rangle. \label{eq:wdw_isolated} \end{equation} Taking the inner product with the scalar field clock bra \langle \varphi \vert{} via the Born rule quadrature map (Definition 1), and applying the Born--Oppenheimer adiabatic approximation (where the universal state factors approximately as \Psi(a, \varphi) \approx \psi(a \vert{} \varphi)\chi(\varphi) with the clock phase varying rapidly compared to scale factor changes), the second-order derivative \partial_\varphi^2 reduces under square-root factorization to a first order Tomonaga--Schwinger evolution equation with respect to \varphi, yielding Eq.~~\eqref{eq:cosmo_schrodinger}. \blacksquare

3. Quantum Gravitational Backreaction and Scalar-Field Clock Fluctuations

Because the scalar field clock \varphi is a quantum operator rather than a classical background clock, it possesses an intrinsic quantum uncertainty \Delta \varphi. Applying our finite-clock dephasing formalism (**Theorem 2**) to minisuperspace cosmology reveals a profound consequence for the early universe: **quantum fluctuations of the scalar-field clock induce intrinsic stochastic blurring of cosmic expansion rates.**

Corollary 2 (Quantum Clock Backreaction and Hubble Variance)

Let the scalar field clock possess a finite quantum uncertainty \Delta \varphi around its mean classical trajectory \varphi(t). The finite clock resolution induces an intrinsic Lindblad-type backreaction on the scale-factor density matrix \rho_S(\varphi), generating an effective variance in the Hubble expansion rate H = \frac{\dot{a}}{a} given by: \begin{equation} \sigma_H^2 = \left(\frac{\partial H}{\partial \varphi}\right)^2 (\Delta \varphi)^2 + \frac{\gamma_{\mathrm{cosmo}}}{(\Delta \varphi)^2}, \label{eq:hubble_variance} \end{equation}

where the first term represents classical trajectory dispersion and the second term is the quantum-gravitational finite-clock correction arising from the double-commutator dephasing term \left[\widehat{H}_{\mathrm{cosmo}}, \left[\widehat{H}_{\mathrm{cosmo}}, \rho_S\right]\right]. As \Delta \varphi \to 0 (infinitely sharp clock limit), backreaction divergences suppress scale-factor off-diagonal coherences, dynamically enforcing decoherence between macroscopically distinct geometry branches (emergent classical spacetime via quantum Darwinism).

Drop-In LaTeX Code for Subsection IX.E (Append to Section IX of the Manuscript)

Paste the following block at the end of \section{Relativistic Spacetime Extension...} in your REVTeX 4.2 .tex file:

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\subsection{Cosmological Minisuperspace: Scalar Field as a Quantum Clock for the Scale Factor}
\label{subsec:minisuperspace}

In canonical quantum cosmology, the Wheeler--DeWitt equation for a flat Friedmann--Robertson--Walker (FRW) universe sourced by a homogeneous scalar field $\varphi$ with potential $V(\varphi)$ is given by~\cite{DeWitt1967,Kuchar1992}
\begin{equation}
   \widehat{H}_{\mathrm{WDW}} |\Psi\rangle = \left[ \frac{1}{2m_0} a^{-p} \frac{\partial}{\partial a} \left( a^p \frac{\partial}{\partial a} \right) - \frac{1}{2a^3} \frac{\partial^2}{\partial \varphi^2} + a^3 V(\varphi) \right] |\Psi(a, \varphi)\rangle = 0,
   \label{eq:wdw_frw_app}
\end{equation}
where $m_0 = 3\pi/(2G)$. Treating the homogeneous scalar field $\varphi$ as the internal quantum clock sector ($H_C = L^2(R)$) and the scale factor $a$ as the system sector ($H_S$), conditioning on a scalar field clock reading via the quadrature map $E(d\varphi) = |\varphi\rangle\langle\varphi|\,d\varphi$ yields the conditional scale factor state $|\psi(a|\varphi)\rangle = \langle\varphi|\Psi\rangle$. Under the adiabatic Born--Oppenheimer expansion, $|\psi(a|\varphi)\rangle$ satisfies the emergent Tomonaga--Schwinger cosmological Schr\"odinger equation:
\begin{equation}
   i \frac{\partial}{\partial \varphi} |\psi(a | \varphi)\rangle = \widehat{H}_{\mathrm{cosmo}}(a, \varphi) |\psi(a | \varphi)\rangle, \qquad \widehat{H}_{\mathrm{cosmo}} = \sqrt{\frac{2m_0}{a^3}} \left(a^3 V(\varphi) - \frac{1}{2m_0} a^{-p}\partial_a(a^p \partial_a)\right).
   \label{eq:cosmo_schrodinger_app}
\end{equation}
Furthermore, finite scalar-field clock uncertainty $\Delta\varphi$ induces quantum gravitational backreaction governed by Theorem~\ref{thm:lindblad}, predicting an intrinsic Hubble expansion variance $\sigma_H^2 = (\partial_\varphi H)^2(\Delta\varphi)^2 + \gamma_{cosmo}/(\Delta\varphi)^2$ that dynamically suppresses macroscopic geometry superpositions and drives emergent classical spacetime.

And add the following references to your bibliography:
\bibitem{Kuchar1992}
K.~V.~Kucha\v{r}, \emph{Time and interpretations of quantum gravity}, in \emph{Proceedings of
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the 4th Canadian Conference on General Relativity and Relativistic Astrophysics}, edited by S.~Kunstatter, D.~E.~Vincent, and J.~G.~Williams (World Scientific, Singapore, 1992), pp.~211--314.